

Nanjing University of Information Science and Technology

Laboratory Report

Experiment Title: Distributed Object Storage & Fault Tolerance Testing

Date	2026-06-12	Class	2
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GitHub Account	whatokk	Repository	whatokk/lab9-distributed-object-storage
Environment	GitHub Actions Ubuntu runner	Storage	MinIO / Docker Compose

一、Experimental Purpose

- Master object storage by understanding bucket/object organization instead of traditional file paths.
- Deploy a distributed MinIO object storage service using Docker Compose.
- Observe metadata and object fragments stored across multiple simulated disks.
- Experience fault tolerance by deleting one storage directory and verifying that data remains readable.

二、Experiment Contents

- Created four independent storage directories: data1, data2, data3, data4.
- Created docker-compose.yml for MinIO server and MinIO Client containers.
- Started cloud-minio and cloud-mc containers.
- Configured mc alias myminio.
- Created bucket my-cloud-storage.
- Generated and uploaded a 20 MiB random object bigfile.bin.
- Observed xl.meta and part fragments across all storage directories.
- Stopped the cluster, deleted and recreated data3, restarted the cluster.
- Verified that bigfile.bin was still accessible and hash-identical after recovery.

- Cleaned up containers and saved experiment evidence.

三、 Detailed Steps and Results

Step 1-2 Deploy Directory Structure

Created a dedicated experiment project and four storage directories to simulate four independent storage disks.

Step 3-4 docker-compose.yml Explanation

The minio service uses quay.io/minio/minio, mounts data1-data4 as volumes, exposes ports 9000 and 9001, sets root credentials, and starts distributed storage with server /data{1...4}. The mc service uses minio/mc and depends on minio so it can manage buckets and objects.

Step 5 Launch Cluster

Ran docker compose up -d. docker ps showed cloud-minio and cloud-mc running, with MinIO ports 9000 and 9001 mapped.

Step 6 Configure MinIO Client Alias

Ran mc alias set myminio http://minio:9000 admin password123. The output showed Added myminio successfully.

Step 7 Create Bucket

Created bucket my-cloud-storage with mc mb.

Step 8 Create and Upload Large File

Generated bigfile.bin using dd with size 20 MiB, copied it to the mc container, and uploaded it to myminio/my-cloud-storage/.

Step 9 Verify Upload

mc ls showed bigfile.bin as a 20MiB STANDARD object.

Step 10 Observe Backend Fragmentation

The backend directories contained xl.meta metadata and part.1/part.2 fragments under data1, data2, data3, and data4. This shows MinIO stored object data and metadata across the simulated disks.

Step 11 Simulate Disk Failure

Stopped the cluster, deleted and recreated data3 with sudo rm -rf data3, then restarted the cluster.

Step 12 Verify Fault Tolerance

After the simulated failure, mc ls still listed bigfile.bin. The object was downloaded as recovered-bigfile.bin and its SHA256 hash matched the original exactly.

Step 13 Cleanup

Ran docker compose down and verified with docker ps that no containers were left running.

四、Evidence Screenshots

Lab9 Experiment Run Summary

Captured on Windows 10 | 2026-06-12 01:25:22 +08:00 | GitHub account: whatokk

Repository: <https://github.com/whatokk/lab9-distributed-object-storage>

Workflow: Lab9 Distributed Object Storage

Run URL: <https://github.com/whatokk/lab9-distributed-object-storage/actions/runs/27364710248>

Status: completed

Conclusion: success

Event: push

Head SHA: bfcc7c68af3dd282775c61dc2546298bf1998e2e

Figure 1. Lab9 successful GitHub Actions run.

4-Disk MinIO Cluster Deployment Evidence

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docker-compose.yml key content:

services:

minio:

image: quay.io/minio/minio

container_name: cloud-minio

volumes:

- ./data1:/data1

- ./data2:/data2

- ./data3:/data3

- ./data4:/data4

ports:

- "9000:9000"

- "9001:9001"

environment:

MINIO_ROOT_USER: admin

MINIO_ROOT_PASSWORD: password123

command: server /data{1...4} --console-address ":9001"

mc:

image: minio/mc

container_name: cloud-mc

depends_on:

- minio

entrypoint: /bin/sh

command: -c "tail -f /dev/null"

tty: true

docker ps after launch:

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS
98a23f3a5762	minio/mc	"/bin/sh -c 'tail -f..."	21 seconds ago	Up 20 seconds	
cloud-mc					
35f04037bf62	quay.io/minio/minio	"/usr/bin/docker-ent..."	21 seconds ago	Up 20 seconds	
0.0.0.0:9000-9001->9000-9001/tcp, [::]:9000-9001->9000-9001/tcp	cloud-minio				

Figure 2. Docker Compose configuration and running MinIO containers.



Figure 3. Bucket creation and 20 MiB object upload.

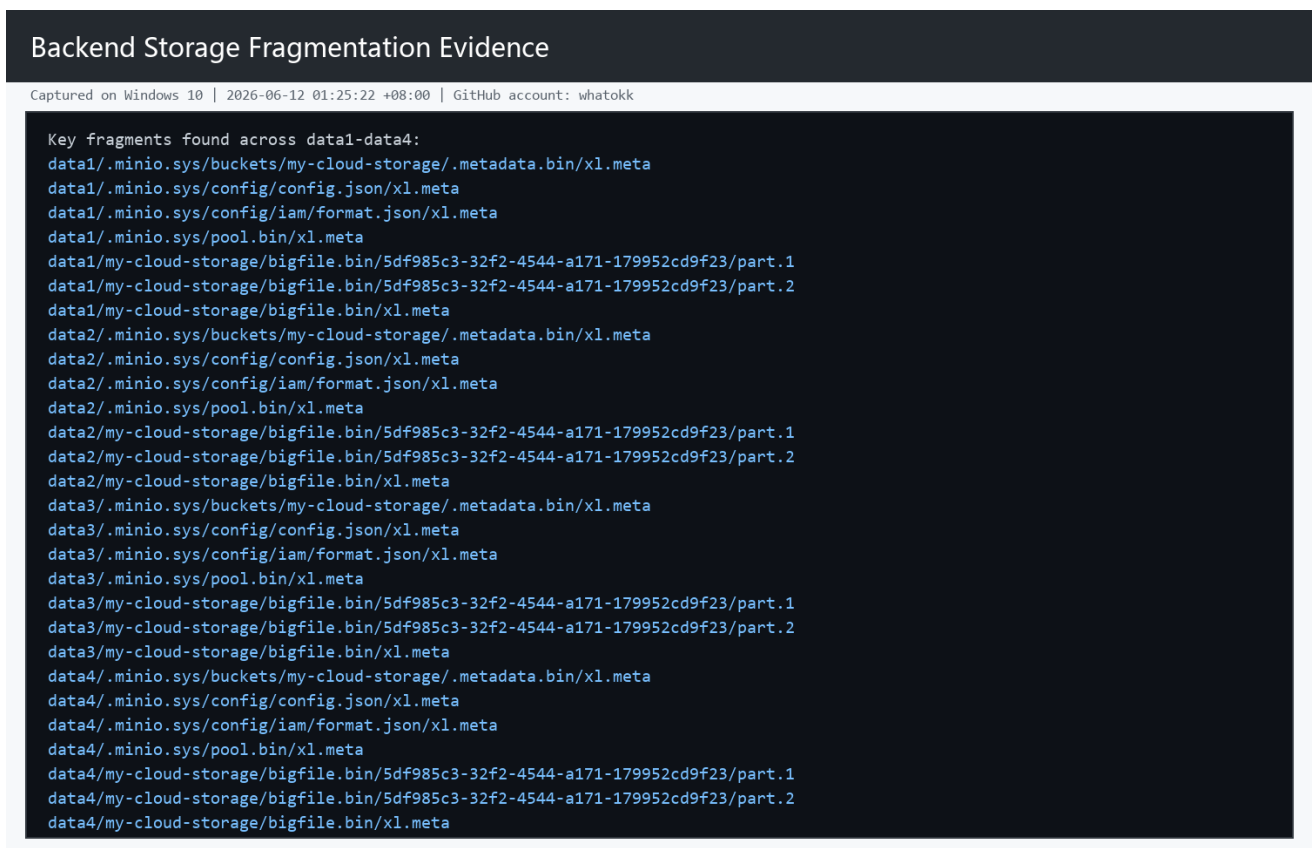


Figure 4. Backend metadata and fragment distribution across four disks.

Disk Failure and Fault Tolerance Evidence

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```
Failure simulation:
data3 deleted and recreated to simulate disk failure
total 8
drwxr-xr-x 2 runner runner 4096 Jun 11 17:19 .
drwxr-xr-x 9 runner runner 4096 Jun 11 17:19 ..

docker ps after restart:
CONTAINER ID   IMAGE      COMMAND                  CREATED        STATUS        PORTS
NAMES
bf92f09e9be9   minio/mc   "/bin/sh -c 'tail -f..." 25 seconds ago Up 25 seconds
cloud-mc
3575d995c46b   quay.io/minio/minio  "/usr/bin/docker-ent..." 25 seconds ago Up 25 seconds
0.0.0.0:9000-9001->9000-9001/tcp, [::]:9000-9001->9000-9001/tcp   cloud-minio

Object listing after deleting data3:
[2026-06-11 17:19:36 UTC]  20MiB STANDARD bigfile.bin

SHA256 comparison:
fac141f56ffa18065dd737647253367f9aecb0018d8456128291a0256a6eda34  bigfile.bin
fac141f56ffa18065dd737647253367f9aecb0018d8456128291a0256a6eda34  recovered-bigfile.bin

Fault tolerance verified: bigfile.bin remained readable after deleting one storage directory.
```

Figure 5. Deleting data3 and verifying fault tolerance.

Cleanup Evidence

Captured on Windows 10 | 2026-06-12 01:25:22 +08:00 | GitHub account: whatokk

CONTAINER ID	IMAGE	COMMAND	CREATED	STATUS	PORTS	NAMES
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Figure 6. Cleanup verification.

五、Reflection

Problems Encountered and Proposed Solutions: The local Windows Docker daemon was unavailable, so the experiment was executed on a GitHub Actions Ubuntu runner. During the first run, deleting data3 failed because MinIO-created files were owned by container users. The command was corrected to `sudo rm -rf data3`, matching the lab instruction and allowing the fault simulation to complete successfully.

Why is data still available after deleting one storage directory? MinIO distributed mode uses erasure coding and stores data plus parity information across multiple disks. The object is not stored as one single file on one disk; instead, metadata and fragments are spread across the four storage

directories. When data3 was removed, the remaining disks still contained enough data and parity fragments to reconstruct the object. This is why bigfile.bin was still readable and its recovered SHA256 hash matched the original.

六、Links

Experiment repository: <https://github.com/whatokk/lab9-distributed-object-storage>

Successful Actions run: <https://github.com/whatokk/lab9-distributed-object-storage/actions/runs/27364710248>